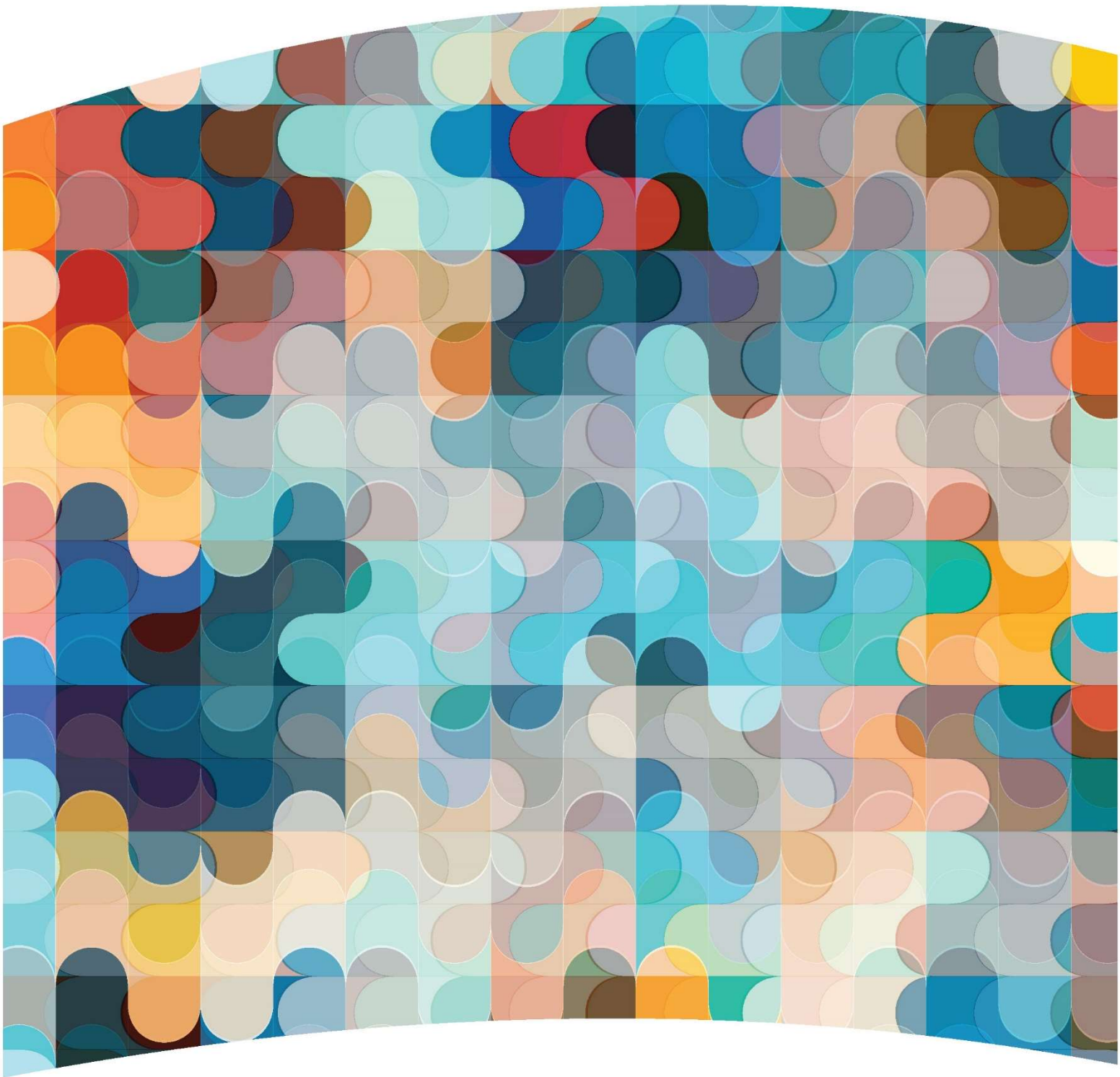
2030 Census IT Strategy and Roadmap

Issued April 2024
Version 1.0



United States® Census Bureau

U.S. Department of Commerce
U.S. CENSUS BUREAU
census.gov

Approval Signature



4/8/2024

Deborah Stempowski
Associate Director for Decennial Census Programs

Date Signed

Version History

The document version history recorded in this section provides the version number, the date it was issued, and a brief description of the changes since the previous release. Baseline releases are also noted.

Version	Date	Description
1.0	03/07/2024	Baselined version of the IT Strategy and Roadmap for public release.

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1 Overview

The *2030 Census Information Technology Strategy and Roadmap* (ITSR) documents the direction and priorities that will be used to achieve the 2030 Census Program. The term information technology (IT) solution refers to the entire assemblage of IT components supporting the mission, goals, objectives, and requirements of the 2030 Census, including system, software, hardware, services, human resources, processes, facilities, trainings, policies, programs, and documentation. The ITSR forms the highest level of IT programmatic policy and informs all derivative IT plans and artifacts for the 2030 Census.

The major tenets of this document are:

- The IT solution will align with and support the 2030 Census business needs.
- The IT solution will be a result of the integration of enterprise systems, purpose-built and contracted systems, and services.
- The 2030 Census IT solution will incorporate the Business Ecosystem as defined in the *Integration Phase One Plan for the Census Bureau's Four Key Initiatives* and *Phase Two Plan for the Census Bureau Business Ecosystem.*
- Major decisions will be data-driven considering multiple factors including security, risk, functionality, and cost.
- The IT program will collaborate closely with partners within the 2030 Census Program and the enterprise.
- A program of research of available technologies will inform the selection of systems that make up the IT solution.
- Acquisitions will be secured to permit exercise of contractor solutions during the 2028 Dress Rehearsal and, if possible, the 2026 Census Test.
- Decennial-specific integration testing will begin early in the program life cycle providing time for iterative development and solution maturation, without duplicating existing testing capabilities in the Business Ecosystem and contractor systems.

2 Introduction

The decennial census requires an accurate count of a highly diverse and growing population projected to exceed 330 million people in more than 140 million housing units (HUs). The goal of the Census Bureau is to account for every person in the nation, in an environment of an increasingly diverse population and complex living situations, exacerbated by increasing concerns about privacy and distrust in the government. Decennial censuses face other challenges including budget constraints, leadership changes, potential continuity disruptions,

and a need to employ hundreds of thousands of temporary workers for a short period of time. A well-designed census can help mitigate the impact of these challenges.

To ensure a successful census, efforts must be initiated early with clear objectives and outcomes defined. Planning for the 2030 Census began in 2019. The 2030 Census Program is being conducted in four phases, beginning with Early Planning (FY19-FY21), followed by Design Selection (FY22-FY24), Development and Integration (FY25-FY29), and Peak Production and Closeout (FY29-FY33). ITSR development began in the Design Selection Phase and is critical to the successful execution of a high-quality census of population and housing. In turn, successful execution is dependent on a strategy that incorporates the latest technology and lessons learned throughout the census life cycle that address the changing environment and inherent logistical challenges of such a large and complex undertaking. Therefore, it is imperative for program success to define IT strategies and a roadmap of key activities and milestones to enable census performance across the remainder of the decade.

This document supports the *2030 Census Operations Strategy and Roadmap*. It is critical that users of the ITSR have read and understood that document.

2.1 Document Purpose

This document articulates the strategies and key milestones for the successful design, development, and implementation of the technical solution to meet the needs of the 2030 Census operational design. The ITSR documents its alignment to key strategic documents within the 2030 Census Program to ensure the IT solution enables and supports the 2030 Census mission.

This document accomplishes the following:

1. Provides insight into the approaches and rationales that the 2030 Census Program leadership will use to design, develop, and operate the IT solution.
2. Serves as a guide for the management of the design, construction, and delivery of the IT solution.
3. Serves as input to the development of the *2030 Census Acquisition Strategy* to deliver systems that will be part of the IT solution.
4. Demonstrates how operational and technical strategies align with the objectives of the 2030 Census Program outlined in the *2030 Operations Strategy and Roadmap*.

The purpose in developing this document in the Design Selection Phase is to provide ample analysis and planning to inform the 2030 Census operational design and requirements. Completing this work earlier in the life cycle allows for more detailed, lower-level system architecture design, and development in the later phases of the life cycle. This document will evolve to incorporate information as it becomes available throughout the program. The 2030

Census Program will update this document, as needed, following critical milestones related to integration and operational testing to incorporate ongoing lessons learned.

2.2 Document Scope

The scope of this document includes the full IT life cycle covering planning, design, development, testing, implementation, execution, and decommissioning. This document describes the strategies and initial plans for the design, development, and integration of the entire 2030 Census IT solution. This includes the integration of internally purpose-built, enterprise, and contracted systems as selected and built through the decade. This document also includes a roadmap of structured implementation activities that provides high-level descriptions, milestones, and key decisions that will guide the development of program level implementation plans.

2.3 Audience

This document is intended to reach Census Bureau staff who plan and implement the census operations and technical staff (internal solution providers, as well as third party contractors), who are responsible for achieving the mission of the 2030 Census. It is also for those interested in obtaining a better understanding of the *2030 Census IT Strategy and Roadmap*. External audiences can use this document as well to have a stake in a successful delivery and execution of the program. Relevant audiences include, but are not limited to the following:

- Individual 2030 Census Program projects including Decennial Senior Leaders, enterprise initiatives, and other key Census Bureau staff (such as current and future system owners and Systems Development Life Cycle [SDLC] teams).
- 2030 Census Program leadership and staff (e.g., 2030 Census Portfolio Management Governance Board members).
- Information technology (IT) chief information officer (CIO) and other Office of the CIO leadership staff.
- Census Bureau division chiefs.
- Department of Commerce (DOC).
- Office of Management and Budget (OMB).
- Government Accountability Office.
- The public.

To set the context for audiences, Section 3.0 introduces the 2030 Census Program including background, vision, mission, and goals of the program.

2.4 Terminology

The terminology used in this, and other program documents, is managed in the *2030 Census Program Architecture Glossary of Terms and Concepts*. These terms are central to using this document as a planning, design, and management tool. It is important that managers have a common understanding and agreement on IT solution terms.

2.5 Document Maintenance

The ITSR is developed, reviewed, baselined, and maintained by the Decennial Information Technology Division using the *2030 Census Document Management Process*.

3 Strategic Alignment

3.1 Derivation of the IT Strategy from the 2030 Census Strategy & 2030 Census Operations Strategy and Roadmap

The *2030 Census IT Strategy and Roadmap* is one of several documents that provide strategic guidance to the 2030 Census Program. The collection of these documents is the authoritative source that details the 2030 Census strategy. These documents and their relationships are depicted in Figure 1.

Figure 1: Derivation of the IT Strategy



At the highest level, the *2030 Census IT Strategy and Roadmap* derives its guidance from the *2030 Census Strategy*. It defines the mission, vision, and goals for the 2030 Census Program, as well as the high-level objectives and strategies for the program. The IT strategy is also influenced by the Census Bureau chief information officer (CIO) strategy in areas that apply to the 2030 Census solution and the associated risk profile.

The *2030 Census Strategy* outlines the program's overall strategic direction and priorities for program planning and definition. It drives all other plans by providing the unifying mission of the 2030 Census, provides a vision for how success is realized over the 2030 Census Program phases, and sets goals, objectives, and strategies that define the mission's success.

The *2030 Census Operations Strategy and Roadmap* describes the approach for developing an evidence-based and integrated operational design that addresses the changing environment and logistical challenges of designing and conducting the decennial census. It describes the iterative operational design process and how that process will be managed. It also includes a

roadmap of key activities and milestones to assist managers in executing the operational design strategies. The *2030 Census IT Strategy and Roadmap* activities contained in Section 4 of this document are derived from the business needs and aligned with and support the processes documented in the *2030 Census Operations Strategy and Roadmap*.

The *2030 Census Acquisition Strategy* describes the approach for acquiring contracts for the 2030 Census. It also outlines key activities and planned enhancements related to the acquisition process.

The *2030 Census Stakeholder Engagement Strategy* will highlight approaches for engagement and activities with oversight entities, expert and advisory groups, and the public for each phase of the decennial census life cycle.

3.2 2030 Census Program IT Strategy Vision/Mission

The decennial census IT vision and mission statements communicate the goals and objectives of the Decennial IT Division (DITD).

VISION: An efficient, effective, and quality census that counts the people of our nation, once, only once, and in the right place.

MISSION: Conduct a census of population and housing and deliver data to the President, the states, and the American people.

3.3 Basis for the IT Strategy

The *2030 Census IT Strategy and Roadmap* documents the direction the 2030 Census Program will pursue to develop and deliver the 2030 Census IT solution. The basis for the IT strategy is derived from the following:

- The *2030 Census Strategy.*
- The *2030 Census Operations Strategy and Roadmap.*
- The *2030 Census Enhancement Area Research Projects.*
- *U.S. Census Bureau CIO Strategy.*
- 2020 Census IT Lessons Learned.

This document considers and aligns with the elements and approaches outlined in those documents including the five guiding principles and along with executive design guidance, assumptions, and constraints.

Figure 2: 2030 Census Operational Mission, Vision, and Guiding Principles



Second, the IT strategy draws from the Census Bureau Office of the Chief Information Officer (OCIO) Strategic Goals which are shown in Figure 3.

Figure 3: Census Bureau OCIO Strategic Goals



Additionally, in establishing this strategy, IT-related lessons learned were applied from previous censuses, with specific focus on the recommendations included in the 2020 Census lessons learned artifact. The following are the top five IT recommendations from the lessons applied:

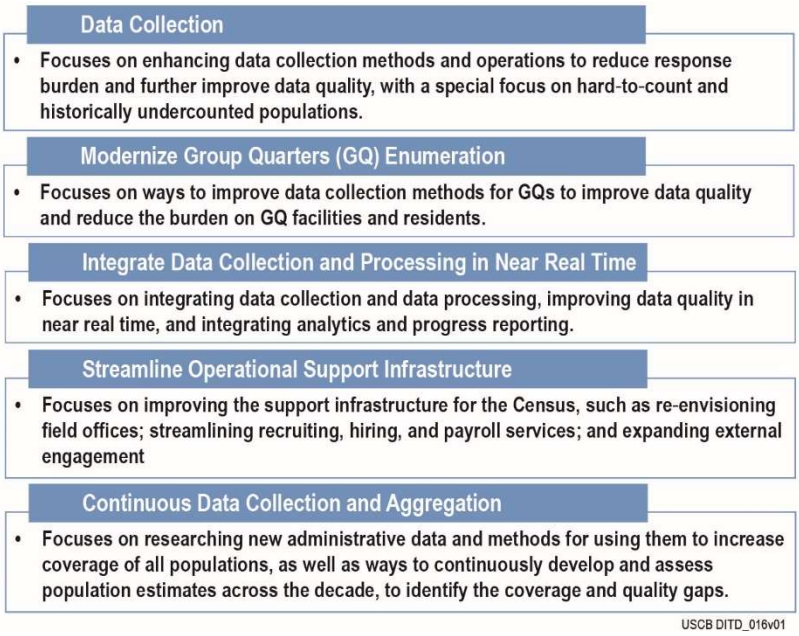
1. *Establish Key Technical Leadership.* Develop Census Bureau staff skills in the DITD and provide the right people at the right time in the right positions. Create clear ownership and responsibilities and enhanced expertise to focus on architecture and integration earlier in the decennial census life cycle.
2. *Establish a Cross Functional Communication Structure*. Leverage the 2020 Census Operational Delivery (OD) construct to drive cross-functional integration and collaboration with teams internal and external to the 2030 Census Program from the start.
3. *Establish a Program Framework.* Create an Integration and Implementation Plan early in the life cycle allowing teams to start early and identify staffing and budget needs. Establishing and enforcing common major milestones enables effective planning for

downstream activities and facilitates earlier and more frequent integration tests maximizing flexibility.

4. *Establish the Program Vision and Processes Early in the Life Cycle.* Establish common processes, scope, tools, and expectations early to ensure consistency and integration across all levels of the 2030 Census Program. The decennial census IT organization must establish ways of working at a small scale to enable efficiencies at a large scale.
5. *Use Governance to Optimize High Performing Teams.* Establish collaborative, efficient, and authoritative governance and explicitly define how decennial and enterprise governance interact. Governance creates a framework for effective decision making using defined structures, the right people, and the right information.

Lastly, the IT strategy supports and aligns with the five Enhancement Areas shown below.

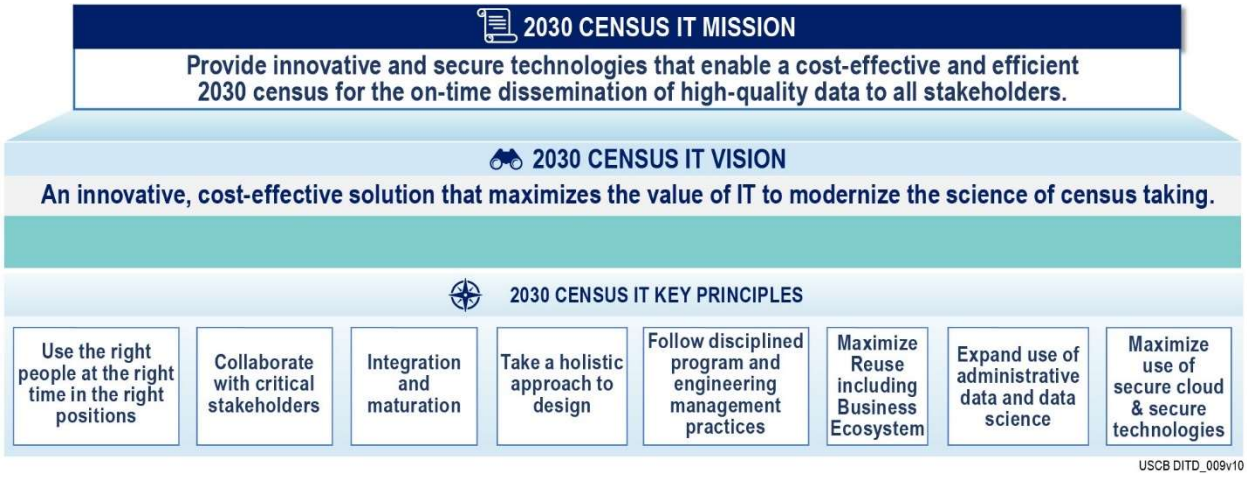
Figure 4: Census Bureau Enhancement Areas



3.4 High-Level IT Strategy

As described above, various strategies, lessons, and goals were leveraged to form this plan such as the *2030 Census IT Mission, Vision, and Guiding Principles* shown in Figure 5 below. These form the high-level elements of the IT strategy that drives the evolution of the people, process, and technology used by the IT organization for the implementation of the 2030 Census solution. Section 4: IT Strategy Implementation Plan further decomposes this strategy as it is applied across the four phases outlined in the *2030 Census Strategy.*

Figure 5: 2030 Census IT Mission, Vision, and Key Principles



The following sections describe each of these IT Key Principles in more detail. These areas are not independent; they work synergistically to maximize the effectiveness of the approach.

3.4.1 Use the Right People at the Right Time in the Right Positions

The key to successful performance is the people, including federal and contractor staff. We need to fill key positions in architecture, engineering, data science, integration, and delivery with the right technical, management and interpersonal skills to maximize solution delivery. Ensuring the organization is effective from the beginning enables the collaboration and integration needed throughout the 2030 Census development and implementation phase of the census life cycle.

3.4.2 Collaborate with Others in the Census Bureau

DITD will drive extensive collaboration with Decennial Census Management Division (DCMD), Decennial Contracts Execution Office (DCEO), and other organizations to coordinate efforts early to reduce implementation risk throughout the 2030 Census life cycle. Building on the successful integration efforts used in the second half of the 2020 Census, we will use integrated teams to collect and understand requirements, inform designs, and status development. This will allow for maximizing the benefit of development while reducing duplication of effort across programs.

As development progresses and solutions undergo iterative improvement, this collaborative mindset and practice will flourish based on its early instantiation. We will learn through the

2030 Census testing events collectively to apply lessons learned consistently across the 2030 Census solution.

3.4.3 Integration and Maturation

The final 2030 Census solution will be the product of continuous, iterative development based on the results of lessons learned and large-scale field tests. This learning environment enhances Census Bureau's ability to capitalize on and integrate emerging technology and mature the solution over time to best meet evolving requirements. The collaborative environment established will provide a place for integrated review of requirements, design, and development to ensure the 2030 Census IT Program continues to look across the enterprise for most beneficial impact and balanced system maturity.

3.4.4 Take a Holistic Approach to Design

While maintaining independence and avoiding tightly coupled systems, 2030 Census architects will view the System of Systems (SoS) holistically to drive systems to support each other. Such an approach results in cost-effective, simple solutions that maximize automation. Architects ensure that downstream systems accommodate potential issues from those upstream and upstream systems minimize issues downstream. There will be particular focus on security and quality as it is imperative that all systems work together and securely early in the life cycle to provide comprehensive solutions with maximized flexibility in areas such as data, application, network, and processes.

Additionally, cost is a key consideration in the IT decision-making process. Processes and solutions will maximize the use of automation as is practical. Examples of this include making solutions scalable to accommodate peak production volumes as well as save cost at lower scales during non-peak times or in lower environments. Finally, simplified designs need to be developed that are flexible and robust. An example of a holistically designed, simplified solution is the elimination of redundant quality systems, the centralization of quality control, and the use of the Business Ecosystem into the 2030 Census IT Solution. This ensures a consistent, efficient approach and provides the potential for an approach that can be used beyond the decennial census.

3.4.5 Follow Disciplined Program and Engineering Management Practices

The 2030 Census IT Program will continue to mature its engineering and program management methods to ensure alignment with 2030 Census Program's Project Management (PM) plans and policies. We must maximize data-driven decision-making and lean, agile approaches leading to predictable, repeatable, consistent results in an environment that requires an ability to handle

change rapidly and efficiently. This will include development of implementation plans to execute IT program management in a manner that aligns with 2030 Census Program PM processes. The areas of focus include risk and issue management, change management, schedule management, contract management, budget/cost management and reporting, business requirements management, IT solution architecture development, IT solution development, and stakeholder engagement.

3.4.6 Maximize Reuse including the Business Ecosystem

The 2030 Census IT Program will reuse artifacts, processes, and systems to increase efficiency with careful attention not to introduce risk. This includes material from the 2020 Census program as well as leveraging agency programs such as the Business Ecosystem. The collaboration between DITD and DCMD, DCEO, and other Census Bureau staff established early in the process at requirements gathering and solution design will enable effective reuse of solutions across the enterprise. Leveraging enterprise programs has many benefits including spreading the costs across the life cycle. This approach also drives more robust solutions both for the decennial census as well as the Enterprise and increases the efficiency with which the Census Bureau provides its services.

3.4.7 Expand Use of Administrative Data and Data Science

The Census Bureau is a data-rich environment allowing for expansive use of administrative data to help achieve the mission of the 2030 Census to account for everyone in the nation once, only once and in the right place. When possible, service providers will expand this use and incorporate machine learning and advanced data science principles. The 2030 Census IT strategy will follow the Census Bureau digital transformation effort to look to more data-centric and data science algorithms and designs to expand the possibilities for potential solutions.

3.4.8 Maximize Use of Secure Cloud and Secure Technologies

First, we will implement a Cloud Smart Strategy. During the 2020 Census, the many benefits of leveraging secure cloud solutions were experienced. The 2030 Census IT strategy will expand on this to maximize those benefits and extend them into other areas of the solution, making it more flexible and cost effective. This includes the use of open-source technology, cloud native development best practices, managed cloud services, such as software as a service and serverless capabilities, and everything as code to automate as much as possible.
Additionally, a resilient and innovative cybersecurity program will be a key ingredient to ensure mission success in the face of an evolving and relentless threat landscape. A robust,

comprehensive defense in depth and cyber intelligence program is critical to safeguard data, systems and interactions against potential threats and vulnerabilities.

3.5 Additional Considerations

In developing the ITSR, the following challenges, critical success factors, and key constraints were also considered.

3.5.1 Security

The 2030 Census solution architecture will align with enterprise objectives for a resilient and innovative cybersecurity program. The architecture and system designs will leverage immutable infrastructure and maximize cloud native and cloud optimized services to maximize automation, artificial intelligence, machine learning and zero trust principles. Data encryption will be a fundamental technique to protect information assets. There shall be a robust, centralized cyber intelligence program that makes use of centralized logging and other techniques to maintain situational awareness. The security approach will also minimize the attack surface and address all applicable threat vectors including those from insider and outsider threats.

3.5.2 Challenges

The table below shows some of the current and future challenges to successful IT strategy implementation.

Current Challenges
Accommodation of emerging, high value technology throughout development and test while balancing risk.
Inconsistent engineering management practices among decennial service providers.
Enterprise systems, commercial off-the-shelf (COTS) products, and their support processes built for long term, low volume workloads vs the short-term, high-volume workloads required by the decennial census.
Future Challenges
Accommodating required updates to systems that also support non-decennial census product operations.
Evolving security adversary skills/techniques requiring an evolution of the security architecture/approach. This continuously evolving threat landscape the solution makes stabilizing the functional and performance aspects of the solution difficult.

3.5.3 Critical Success Factors

The following are factors that are critical to the success of the implementation of the ITSR:

Critical Success Factors	Rationale
Sufficient funding and resources will be available to staff engineering efforts.	Budgets and lack of resources will not be a prohibitive factor in terms of effectively planning and implementing this strategy.
The enterprise IT organization will collaborate with 2030 Census teams.	A key to successful application of enterprise solutions will be the partnership and collaboration established with the Census Bureau's Office of the Chief Information Officer (OCIO). This will ensure that the enterprise solutions satisfy Decennial requirements within the 2030 Census Program's schedule.
The business and IT Solutions organizations will collaborate with 2030 Census teams.	A major key to the solution teams understanding and meeting the needs of 2030 Census is to have a high level of collaboration and communication with DCMD.
Enterprise solutions, including BE, will be built for at least Minimal Viable Product and be able to support the 2030 Census Program.	Base Enterprise solutions will support Decennial needs. Changes for Decennial-specific function will not require major rework and put schedules at risk.
Acquisition timelines will support the needs of the 2030 Census Program.	The acquisition process can be lengthy. It is important for IT and acquisition teams to work closely together to fully understand the acquisition and requisite processes and approvals needed to award contracts in time to support census tests and the 2030 Census.
Census Bureau staff will participate in requirements development and resolve requirement/spec requirement issues in a timely manner.	A major key to the solution teams understanding and meeting the needs of 2030 Census is to receive clear and timely requirements from DCMD.
Research and Methodology organizations will collaborate on development of algorithms that facilitate more efficient approaches to data processing.	Some legacy algorithms were constructed in a way that assumed batch or "end of collection" processing. These may require refactoring to allow for more transactional/real-time processing approaches.

3.5.4 Key Constraints

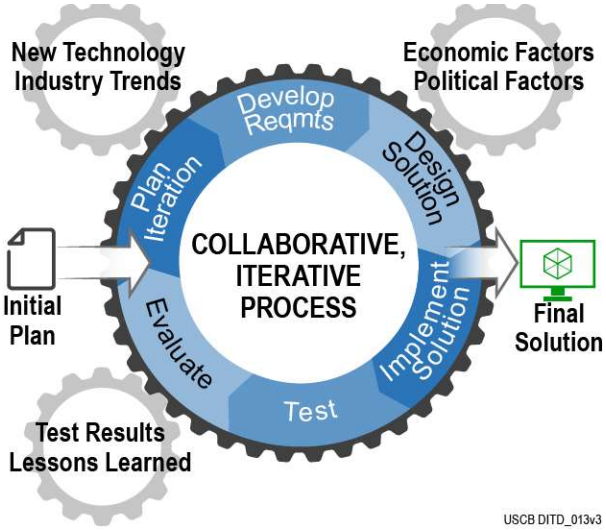
The following are key IT constraints that drive aspects of the 2030 Census IT strategy:

- Solution must be production ready in time to support data products mandated by law.
- Budget funding uncertainty or unknowns may potentially constrain system of system resource allocations and/or potential risk mitigation/contingencies within the budget framework.

4 IT Strategy Implementation Plan

This section describes the plan for implementing the IT strategy. It discusses how the strategy and key principles discussed in the prior sections are applied throughout the four phases of the 2030 Census Program's life cycle: Early Planning, Design Selection, Development and Integration, and Peak Production and Close-Out.

Figure 6. Iterative Process Model

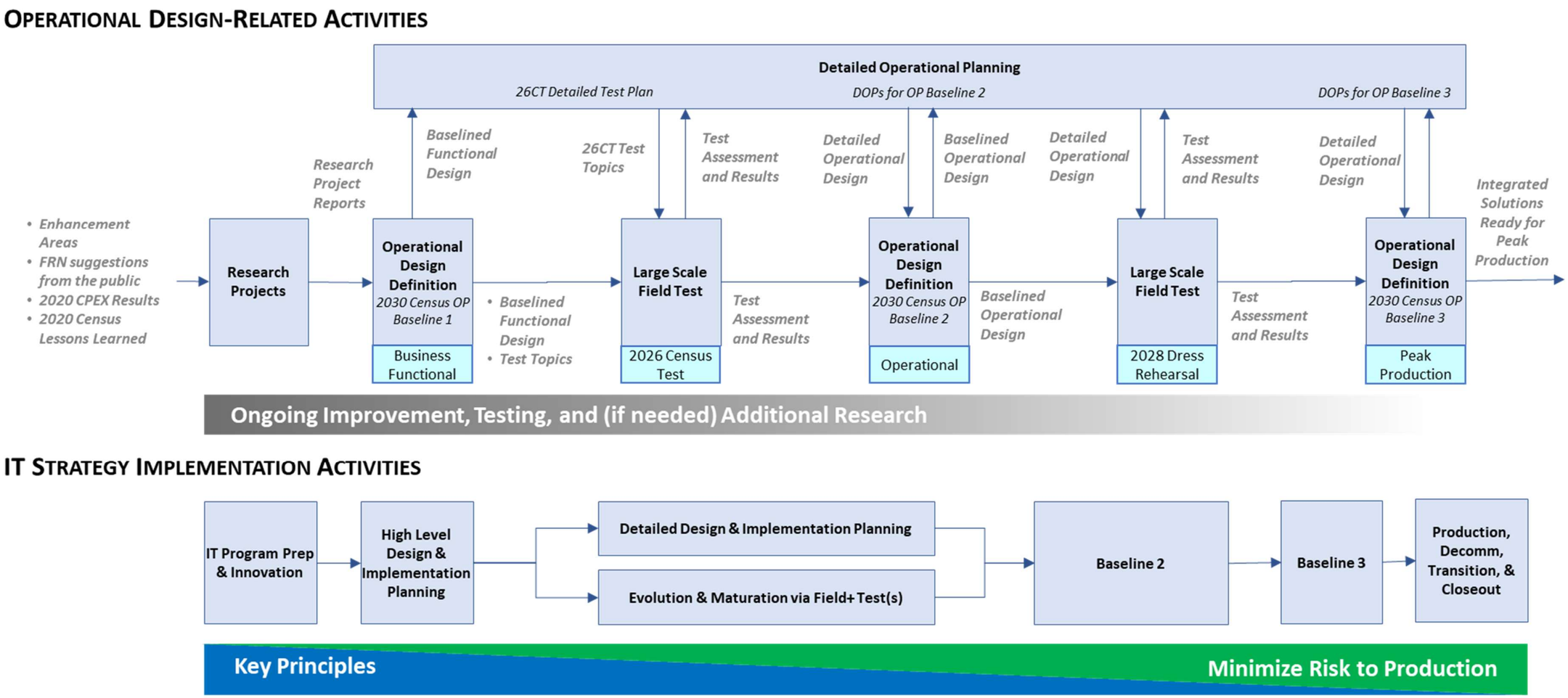


A key overall theme to the IT strategy is the use of an Iterative Process Model to mature the solution over time. As elements of the higher-level plan are applied to each of the phases, they are broken down into more and more detail. In the decennial census environment, it is critical to have agile, flexible processes and designs to accommodate the many factors that could drive change such as emerging technology, lessons learned from tests, and social, geographical, and environmental changes. Given these parameters, this strategy incorporates multiple iterations allowing for changes throughout the life cycle.

4.1 High Level IT Implementation Activities and Alignment with Operational Design Development Activities

A major goal of the ITSR is to align with and support the *2030 Census Operations Strategy and Roadmap*. The diagram below shows the operational design implementation strategy from the *2030 Census Operations Strategy and Roadmap* in parallel with the IT Strategy Implementation Activities. Figure 7 depicts the alignment of these activities.

Figure 7: Operations Design & IT Strategy Implementation Activities



In the IT Implementation, each activity corresponds to and supports a similarly focused operational activity. The technical solutions developed are incorporated into the four baselines called out in the *2030 Census Operations Strategy and Roadmap*. To ensure alignment, throughout the approach there is a high level of collaboration, communication, and coordination with the Decennial Census Management Division.

4.2 IT Program Preparation and Innovation

There are two specific elements within this activity: IT Program Preparation and Innovation. In IT Program Preparation the focus is on the setup of the organizational and programmatic aspects of the effort, such as the overall approach for Systems Engineering Management and the high-level plan for Integration and Implementation. In Innovation (Research and Improvement) the effort focuses on the preparation required for the technical solution.

4.2.1 IT Program Preparation

This area focuses on ensuring that the people, processes, and technology of the organization are in the proper positions for work performance. Efforts include formulation of leadership and organizational structures, maturation of necessary skills, and development and baseline of program frameworks and practices. Additionally, each of these efforts leverage data-driven methods to facilitate real time insights, drive accountability, and maximize transparency.

In this early phase we will baseline high level plans such as:

- Systems Engineering Management Plan (SEMP).
- 2030 Census IT Solution Architecture Vision (Preliminary).
- Integration and Implementation Plan and associated milestone timelines.
- 2030 Census Operational Delivery Management Plan (OD Plan). [1]

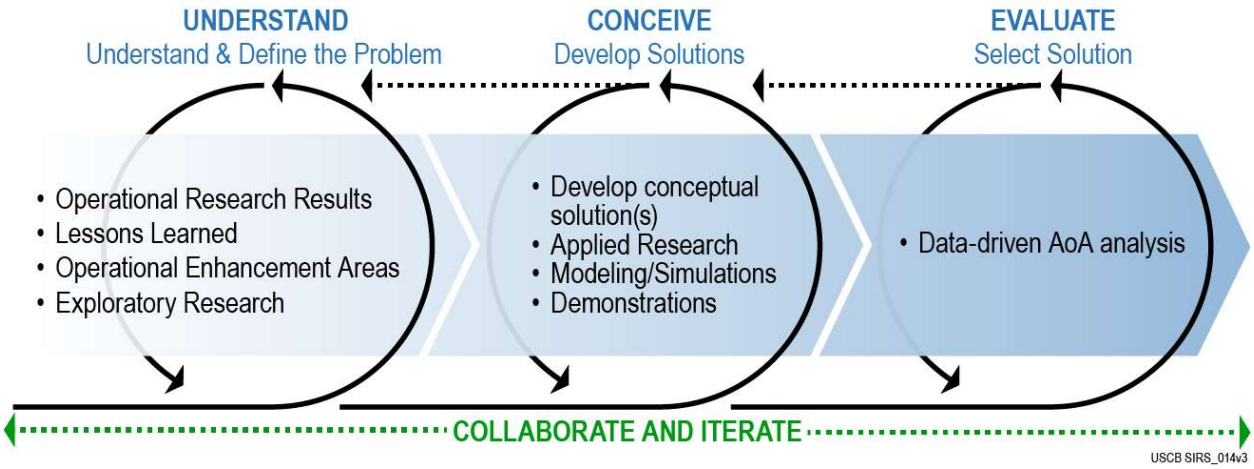
These and other documents will align with higher level strategies and approaches, including those for core IT capabilities provided by the BE, and will provide specific guidance to project and integration teams.

[1] Designation and use of the term “Operational Delivery Team” (OD) may shift following discussions and decision by DCMD after document publication.

4.2.2 Innovation (Research and Improvement)

The basic approach to this activity includes three primary steps as shown in Figure 8 below. Each of these steps is iterative in nature. Time to complete can vary greatly depending on the subject matter.

Figure 8: Approach to Designing Solutions for Large, Complex Problems



4.2.2.1 Understand

The key goals of this step are to understand and define a problem and its context. In some cases, problems are clearly understood and/or previously identified by lessons learned. For these, the approach requires a minimal or moderate amount of collaboration with subject matter experts to define a solution. In other cases, little is known about the problem itself or potential solutions. This requires a more extensive amount of collaboration and exploratory research.

4.2.2.2 Conceive

This activity is performed to develop conceptual or prototype candidate solutions. This can be as simple as a single COTS product choice or as complicated as a redesigned collection of systems and product choices. The proposed solution design provides enough fidelity for evaluation and recommendation. The following is an initial list of the areas of applied research:

- Systems Engineering.
- 2030 Census Solution Common Services Areas.
- 2030 Census SoS Architecture Areas.

This list of research will be more specifically defined, prioritized, and executed throughout the IT Program Preparation activity leading to the incorporation of additional topics.

4.2.2.3 Evaluate

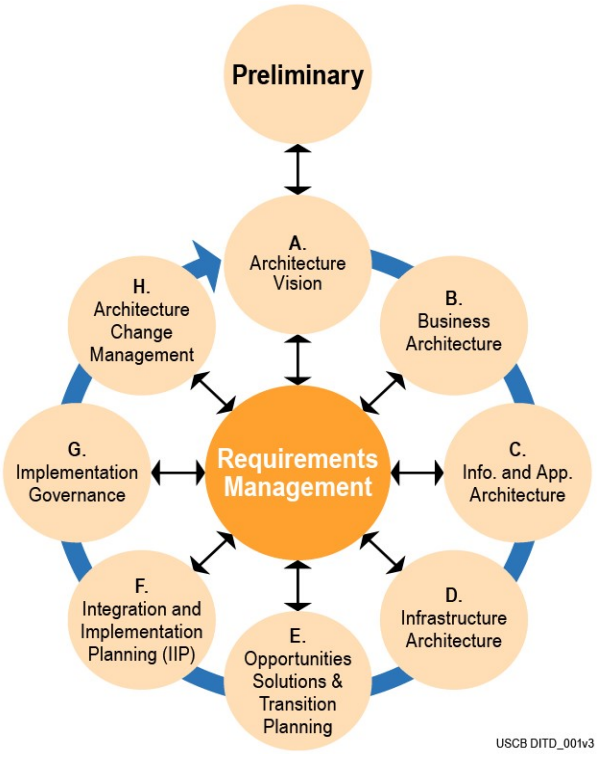
A data-driven method such as an Analysis of Alternatives (AoA) will be used to select a solution from those evaluated. The AoA includes a set of defined, weighted criteria applied to each candidate solution allowing for accurate comparison. At least five key parameters including total cost of ownership, maturity, subsequent applicability to the Census Bureau enterprise, scalability, and security will be assessed. These results, assessments, and recommendations then progress through the appropriate IT governance for implementation.

4.3 High Level Design and Implementation Planning Activity

4.3.1 Methods

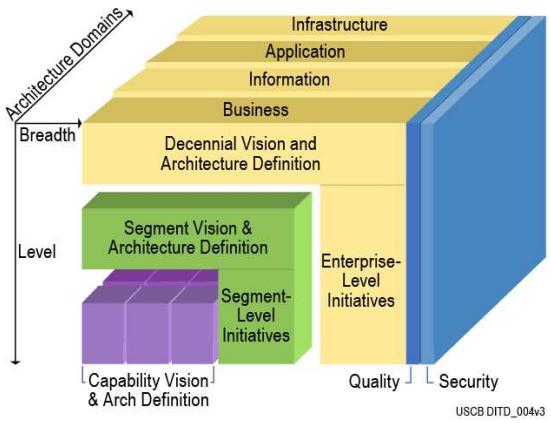
The approach to developing the 2030 Census System of Systems (SoS) Architecture is based on The Open Group Architecture Framework (TOGAF) Architecture Development Method (ADM). We have also tailored the TOGAF approach using elements of the Zachman Framework to organize the various artifacts. The ADM starts with an architecture vision that achieves the strategic elements of the IT strategy and incorporates the results of research and improvement activities. Following the vision, each of the domains for the target architecture as shown in Figure 9 is created.

Figure 9: 2030 IT Architecture Development Model (ADM)



The 2030 Census Program Architecture is a collection of documents and artifacts that provides a complete knowledgebase describing what, how, where, who, when, and why of the 2030 Census Program. The architecture will be responsive to the business requirements of both the 2030 Census operational design and the field tests once those are defined. It starts by describing these aspects of the solution at a high-level and decomposing them to multiple layers of detail. Figure 10 shows the relationship between the various architecture domains and how they are decomposed to finer levels toward implementation. It includes the following architecture domains:

Figure 10. Relationship and Decomposition of Architecture Domains



- Business Architecture: Describes how the solution needs to operate and the flow of work to achieve the business goals and respond to the strategic drivers set out in the Architecture Vision.
- Information Architecture: A description of the structure and interaction of the solution's major types and sources of data, logical data assets, physical data assets, and data management resources.
- Application Architecture: A partitioning of the high-level solution into lower levels which describes structure and interaction of the applications as groups of capabilities or systems that provide key business functions and manage the data assets.
- Infrastructure Architecture: Describes the high-level physical architecture including geographic info, network design, and high-level approach to the common infrastructure provided to the systems.
- Quality Architecture: Embedded in each of the other domains, this domain/view drives a centralized approach to processing quality.
- Security Architecture: Embedded in each of the other domains this domain/view drives an increased focus and coordinated approach to security across the entire solution.

Solution Architecture Transition Plan. At a high level, the *2030 Census Operations Strategy and Roadmap* calls for four operational design baselines:

- Baseline 1: 2030 Census Business Functions and informs the 2026 Census Test Topics.
- Baseline 2: 2030 Census Operations and informs the 2028 Dress Rehearsal Test Topics.
- Baseline 3: 2030 Census Operations for Peak Production and Closeout Phase (Target Architecture).
- Baseline 4: Documents the final "As Was" or "Actual" design, including any adjustments made during 2030 Census production, to inform early 2040 Census planning.

The Solution Architecture Transition plan defines the systems architectures for the first three of these baselines. It governs the evolution of the systems architecture from an initial baseline, through the various field tests, to the target architecture that is to be used for 2030 Census production. Because the engineering approach is iterative and incremental, the solution will evolve and mature during the implementation of each baseline.

4.3.2 Solution Architecture and Methodology Principles

The Architecture and Methodology Principles below flow from the highest-level IT Principles established in Section 3.4. These items align with the Key Concepts, Enhancement Areas, and Executive Design Guidance from the *2030 Census Operations Strategy and Roadmap*. These are critical in setting the foundation for the System of System Solution Architecture and IT Methodologies and Governance:

1. **Security First:** Prioritize security across the architecture to safeguard data, systems, and interactions against potential threats and vulnerabilities.
2. **Interoperability:** Enable seamless communication and data exchange between systems to ensure they can work together effectively.
3. **Resilience and Fault Tolerance:** Design systems resilient to failures, ensuring uninterrupted operation even during disruptions or faults.
4. **Modularity and Reusability:** Build modular components for easier maintenance, scalability, and reuse across different parts of the architecture.
5. **Loose Coupling:** Reduce dependencies between systems to enable flexibility and independent evolution.
6. **Adaptive Security Measures:** Implement dynamic security measures capable of adjusting to emerging threats and evolving environments.
7. **Standardization of Components:** Use standardized technologies and frameworks for simplified integration and reduced complexity.
8. **Service-Oriented Architecture (SOA):** Adopt an architecture with loosely coupled services communicating via well-defined Assisted Person Interviewing.
9. **Data Consistency and Integrity:** Ensure consistency and integrity of data across interconnected systems.
10. **Scalability Across Systems:** Design systems for horizontal and vertical scalability to handle increased workloads.
11. **Cross-System Monitoring and Observability:** Implement comprehensive monitoring to track performance and interactions between systems.
12. **Ethical Considerations:** Incorporate ethical considerations into system design and data usage.

13. **Optimization for Performance and Efficiency:** Continuously optimize for performance using caching, load balancing, and resource optimization.
14. **Regulatory Compliance Across Systems:** Ensure compliance with relevant regulations across interconnected systems.
15. **Real-time Responsiveness:** Enable systems to respond in real-time to events and changing conditions.
16. **Adaptability to Change:** Design systems to be adaptable to new technologies and evolving business needs.
17. **Business Alignment:** Align architecture closely with business objectives to support overall business goals and strategies.

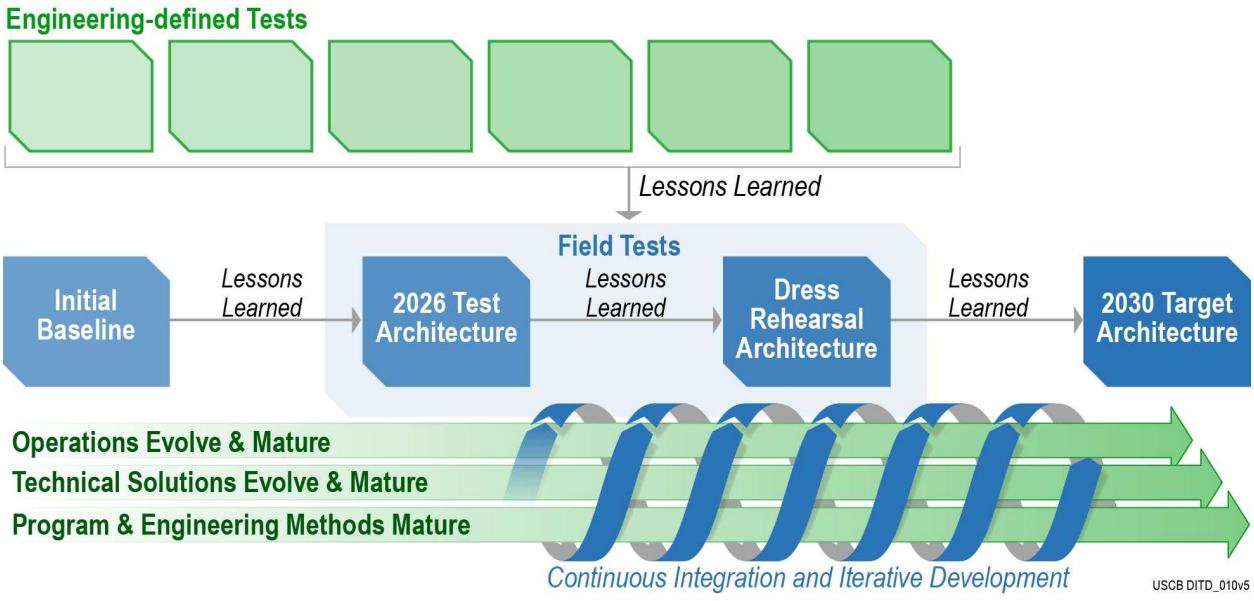
4.4 Detailed Design and Implementation Planning Activity

The *Integration and Implementation Plan* will describe the detailed framework of milestones, decision points, and governance for integration and release of the IT solution and its progression through each of the four baselines. This effort is the detailed planning activity for the subsequent implementation activities.

At its lowest level, the major activities of the 2030 Census IT program will be a process of continuous development. To facilitate integration of the systems, a detailed planning framework will be established along with a timetable of major IT operational deliveries. Each major baseline will promote identified IT capabilities through a series of readiness reviews.

4.5 Evolution and Maturation via Field Test(s) Activity

Figure 11: Maturation of the SoS Architecture through Iterative Testing



This activity occurs as decennial operations and IT solutions evolve and mature. From an IT perspective, after the creation of an initial architecture the solution evolves to the target architecture as shown in Figure 11. This is done to transition the architecture to its target state while supporting each of the field tests. Additional tests may also be scheduled if the field tests only exercise a subset of functionality or if information is needed about a solution on a different timeline. The project level software development, execution, and results of all tests contributes to the progressive maturation of the solution.

4.6 Baseline 2 Activity

This activity includes the creation of the solution baseline that will be used to support the 2028 Dress Rehearsal. This baseline incorporates lessons learned from all earlier field and engineering tests. The key goal of this baseline is to be as close to the 2030 Census production baseline (Baseline 3) as practical.

4.7 Baseline 3 Activity

This activity includes the creation of the solution baseline that will be used to support 2030 Census Peak Production and Closeout Phase (Baseline 3). This baseline resolves any remaining technical debt in the solution and addresses all requirements. All support processes are developed and matured by this phase. Development throughout this phase focuses on ensuring that the system is robust and fully tested in areas of functionality, exception handling, performance, security, and scalability. The major goal of this activity is to reduce production risk.

4.8 Peak Production, Decommissioning, and Closeout Activity

This activity includes preproduction preparation, software launch, peak production operations as well as decommissioning, or transition of systems to other uses. Preproduction includes installation and checkout of solutions and various preparatory testing such as the Operational Test and Dry Run (OTDR). Production operations also include “soft launches” to enable the scale up process to proceed with fewer issues and higher confidence. The solution will have built-in flexibility to adjust during production in response to unexpected challenges. Any changes realized during production will inform “Baseline 4” of the operational design. After production, systems will transition to decommission, downsizing, or reuse on another platform.

5 IT Strategy Roadmap: Major Activities and Milestones

As previously described, a major goal of the *2030 Census IT Strategy and Roadmap* is to align with and support the *2030 Census Operations Strategy and Roadmap*. Figure 12 below shows the operational design implementation strategy roadmap from the *2030 Census Operations Strategy and Roadmap* to provide context for Figure 13 that shows the IT implementation strategy roadmap. Depicting the two processes and timelines this way shows how the IT implementation strategy directly aligns with the Operation Design Strategy.

Figure 12. Operations Roadmap

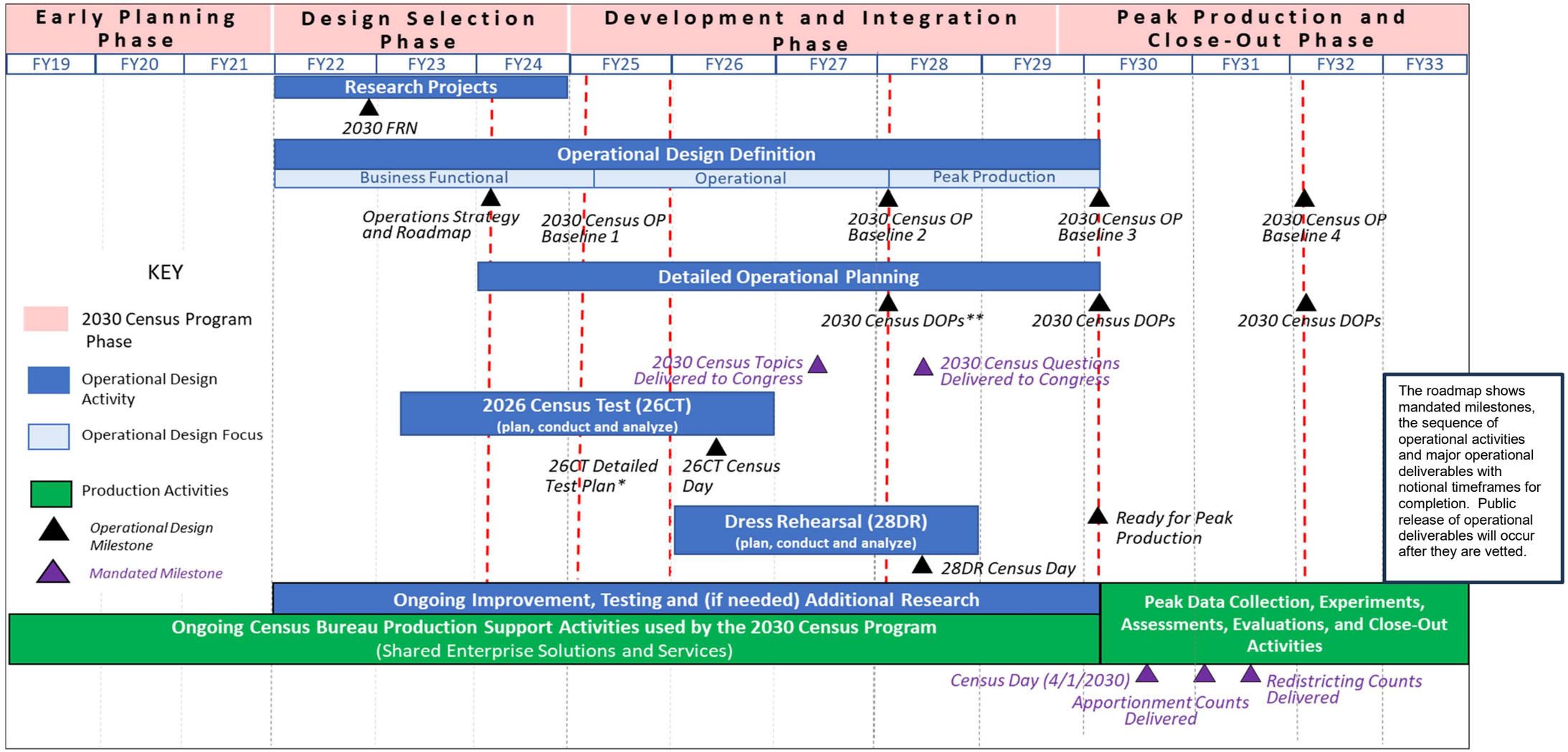
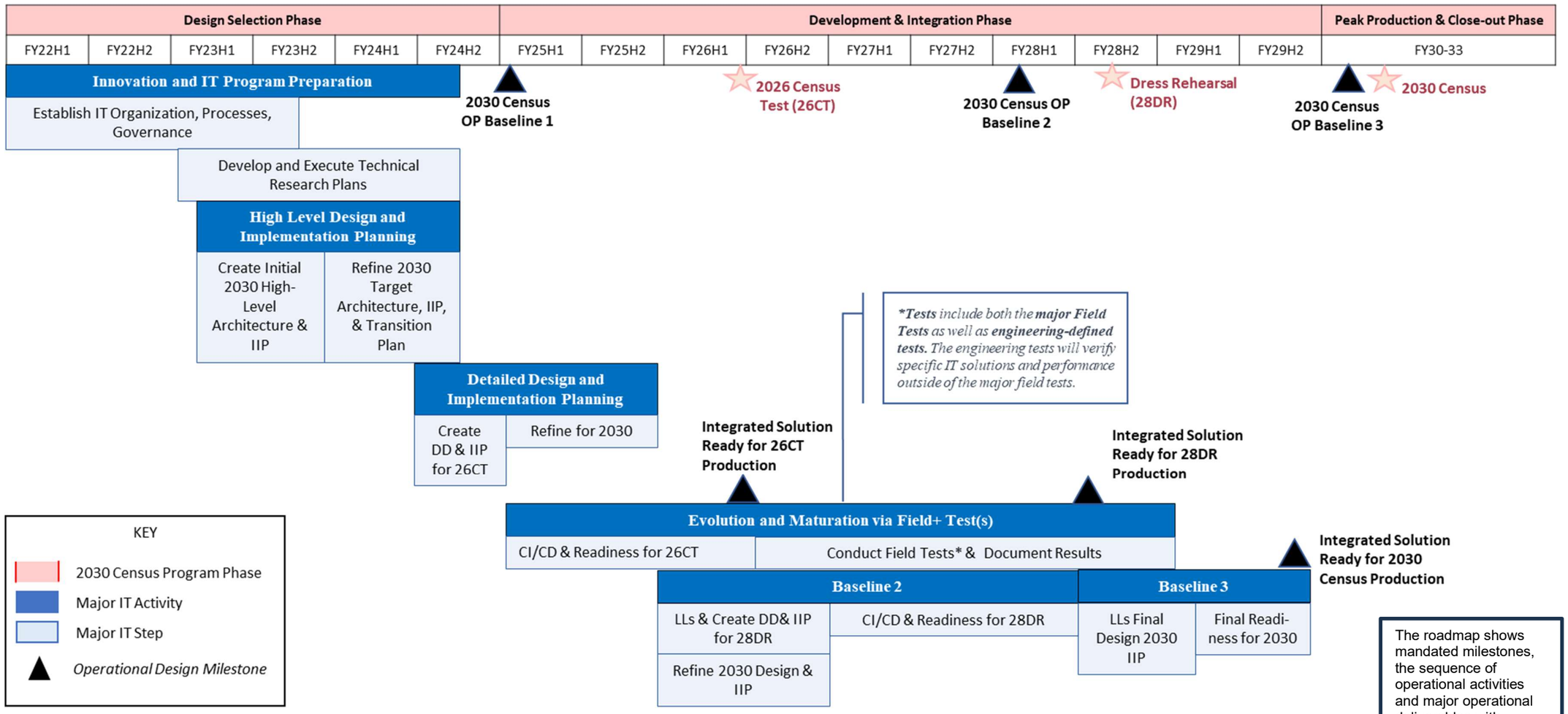


Figure 13. IT Roadmap



FY##H##: Indicates the first or second half of the Fiscal Year

This IT roadmap depicts the iterative nature of the 2030 Census IT strategy. While there is a significant focus on iteration via the three operational design baselines and two major field tests, it is important to recognize the significance of the engineering-defined tests that evolve and mature the 2030 Census solution as well. These 2030 Census-specific tests will validate performance and scalability and specific IT solutions throughout the Development and Integration Phase outside of the major field tests. It should also be noted that this IT roadmap represents an executive-level view of the strategy. The *2030 Census Integration and Implementation Plan* timeline will provide the next level of schedule detail and be included in the 2030 Census Integrated Master Schedule (IMS).

6 IT Strategy Management

In implementing this strategy, the program adheres to a structured governance cadence of change gates and readiness reviews as referenced in Section 4.4. Supporting this series of reviews are governance boards, plans, and metrics that determine how the IT service providers will interact with its stakeholders and report on implementation success.

6.1 IT Governance Boards and Plans

The 2030 Census Program understands that the formation and execution of governance is essential to successful integration. The 2030 Census technical integration governance will be aligned to the Decennial Census Governance and Decision-Making Plan. The implementation plan will include:

- Technical governance board's scope, roles, and responsibilities.
- Interaction and collaboration with exiting IT governance boards and processes (including, but not limited to, Business Ecosystem and OCIO boards and processes).
- Empowerment steps/training for board members to enable cross-functional decision-making.
- Periodic retrospectives on the effectiveness of the technical decision-making process.

The specific technical boards required to govern the 2030 Census technical integration will change as the decennial life cycle progresses. The interactions between related boards (especially enterprise boards) will change as well. The 2030 Census IT Program will continue its iterative approach in the creation and maintenance of boards through memorandums of agreement (MOAs). These will ensure the 2030 Census Program and its solution and service providers work together to plan the support required to meet decennial objectives.

6.2 Metrics

To measure the effectiveness of implementing the ITSR, key performance indicators for each of the key principles discussed in Section 3.3 will be developed. These will be iterative in nature and appropriate to the phase of development.

7 Publicly Available Referenced Documents

Document Name	Version	Date
2030 Census Operations Strategy and Roadmap	V1.0	April 2024
2030 Census Acquisition Strategy	V1.0	April 2024
Census Bureau Strategic Plan—Fiscal Year 2022 Through Fiscal Year 2026		01/31/2022
DOC Strategic Plan 2022-2026		
2030 Census Strategy	V1.0	April 2024
Census Bureau CIO Strategy		01/31/2022
Phase One Plan for the Census Bureau's Four Key Initiatives	V1.0	04/15/2022
Phase Two Plan for the Census Bureau Business Ecosystem	V1.0	07/28/2023

8 Conclusion

This document articulates the strategies and key milestones for the successful design, development, and implementation of the technical solution needed for 2030 Census Program. It also aligns with the key 2030 Census strategic documents, operational design, and requirements. Successfully adhering to this strategy will help achieve the 2030 Census mission.

Appendix: List of Acronyms and Abbreviations

Acronym	Definition
A:	
ADM	Architecture Development Model
API	Assisted Person Interviewing
B:	
BE	Business Ecosystem
C:	
CIO	Chief Information Officer
COTS	Commercial Off The Shelf
D:	
DCEO	Decennial Contracts Execution Office
DCMD	Decennial Census Management Division
DDSSO	Decennial Directorate Support Services Office
DOC	Department of Commerce
E:	
F:	
FY	Fiscal Year
G:	
H:	
HR/HRD	Housing Units
I:	
IIP	Integration & Implementation Plan
IMS	Integrated Master Schedule
IT	Information Technology
ITSR	Information Technology Strategy Roadmap
J:	

Acronym	Definition
K:	
KPI	Key Performance Indicators
L:	
M:	
MOA/MOAs	Memorandum of Agreement/Memoranda of Agreement
N:	

Acronym	Definition
O:	
OCIO	Office of Inspector General
OD	Operational Delivery
OMB	Office of Management and Budget
OTDR	Operational Test and Dry Run
P:	
PM	Project Management
Q:	
R:	
S:	
SDLC	Systems Development Life cycle
SEMP	Systems Engineering Management Plan
SOA	Service-Oriented Architecture
T:	
TOGAF	The Open Group Architecture Framework
U:	
V:	
W:	
X:	
Y:	
Z:	